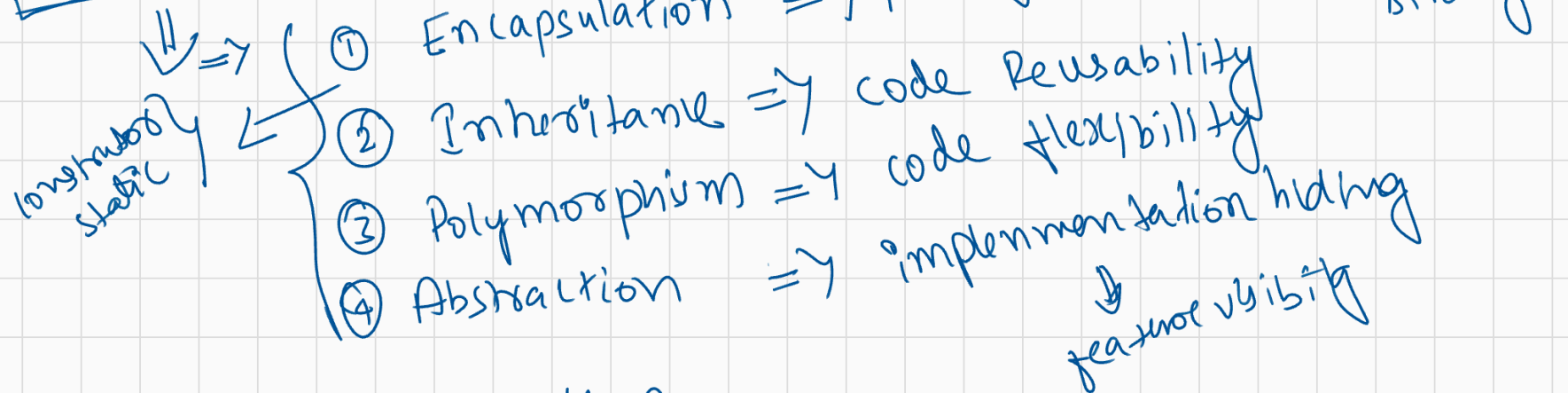


⇒ OOPs :-



③ weeks

① ⇒ Encapsulation



- ① Data hiding ✓
- ② Data Binding ✓
- ③ providing security ✓
- ④ providing controlled access to Data members ✓

private, setter, getter, this keyword, shadowing

Encapsulation

How to achieve encapsulation in java

by using private, getters and setter; if a class have only private data members than its called as bean

Data hiding/binding

providing security

providing controlled access to data members

```
class Student
{
    int age;        //instance variable//Data Member
    String name;    //properties
    String city;

}

//
}

public class LaunchEncap {

    public static void main(String[] args) {

        Student st=new Student();
        st.age=-28;
        st.name="Xp12XP";
        st.city="Bengaluru";
    }

}
```

at the current moment photos shown above anybody can get the object data member and can change it there is no security

To avoid these we have to use keyword private

Private is a keyword which ensures that no one can access or change outside the class

```
class getData
{
    private int data;
    private String name;

}

public class encap {
    public static void main(String[] args) {
        getData gd=new getData();
        // gd.data=12; error will be there as private keywords is used
        //we cant fetch it from outside of the class
    }
}
```

Now to get the data after using the private keywords we can use getter and setter

Setter

if a method is doing an activity of receiving data members from outside and setting its value to its data members such method is called a setter

For a setter some syntax is recommended

recommendation

- please give name set+Var name as method name

Getter

if a method is doing an activity of transferring data members from inside of the class to outside class technically we call it as getter

recomendation

give get+datamember name and getter must not accept anyinput;

```
class getData
{
    private int data;
    private String name;
    //setting setter;
    void setData(int data,String name)
    {
        this.data=data;
        this.name=name;
    }
    int getData()
    {
        return data;
    }
}

public class encap {
    public static void main(String[] args) {
        getData gd=new getData();
        // gd.data=12; error will be there as private keywords is used
        //we cant fetch it from outside of the class
        gd.setData(12,"soham");
        int num=gd.getData();
    }
}
```

```
}  
}
```

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Q>

Given

```
3. public class Spock {  
4.     public static void main(String[] args) {  
5.         int mask = 0;  
6.         int count = 0;  
7.         if( ((5<7) || (++count < 10)) | mask++ < 10 ) mask = mask + 1;  
8.         if( (6 > 8) ^ false) mask = mask + 10;  
9.         if(!(mask > 1) && ++count > 1) mask = mask + 100;  
10.        System.out.println(mask + " " + count);  
11.    }  
12. }
```

Which two are true about the value of mask and the value of count at line 10?
(Choose two.)

- A. mask is 0
- B. mask is 1
- C. mask is 2
- D. mask is 10
- E. mask is greater than 10
- F. count is 0
- G. count is greater than 0

count = 0

mask = 2

Answer: C,F

Which of the following declarations are invalid?

- 1) int[] a=new int[];
- 2) int[][] a=new int[3][4];
- 3) int[][] a=new int[3][];
- 4) int[][] a=new int[][4];
- 5) int[][][] a=new int[3][4][5];
- 6) int[][][] a=new int[3][4][];
- 7) int[][][] a=new int[3][][5];

Answer: 1,4,7(invalid: beocz of dimension missing)

What is the output of this code?

```
int[] a=new int[3];  
System.out.println(a);  
System.out.println(a[0]);
```

- A. CE
- B. [I@...
0
- C. Some problem at the runtime
- D. [I@..
null

Answer: B

What is the output of this code?

```
int[][] a =new int[3][2];  
System.out.println(a);
```

```
System.out.println(a[0]);
System.out.println(a[0][0]);
```

- A. CE
- B. [I@...
[[I@...
0
- C. Some problem at the runtime
- D. [[I@..
[I@...
0
- E. D. [[I@..
[I@...
null

Answer: D

Predict the output

```
int[][] a=new int[2][];
System.out.println(a);
System.out.println(a[0]);
System.out.println(a[0][0]);
```

- A. Compile time error
- B. 0
[I@...
0
- C. [[I@...
null
0
- D. [[I@...
[I@...
0
- E.C. [[I@...
null
NullPointerException

Answer: E

For the above code predict how many objects are created and how many are eligible for garbage collection?

```
int[][] a=new int[3][2];
a[0]=new int[3];
a[1]=new int[4];
a=new int[4][3];
```

- A. object created = 3
eligible for gc = 3
- B. object created = 11
eligible for gc = 6
- C. object created = 10
eligible for gc = 5
- D. object created = 10
eligible for gc = 10
- E. None of the above

Answer: B

What is the nature of the following code?

```
class Test {  
    int[] a; //instance variable -> memory will be created during object creation  
    and default value by JVM  
    public static void main(String[] args) {  
        Test t1=new Test();  
        System.out.println(t1.a);  
        System.out.println(t1.a[0]);  
    }  
}
```

- A. Compile Time Error
- B. 0
- C. [I@...
0
- D. null
- E. null
NullPointerException

Answer: E

Q> Here a is declared at instance level

```
int[] a=new int[3];  
System.out.println(obj.a);  
System.out.println(obj.a[0])
```

- A. Compile Time Error
- B. 0
- C. [I@...
0
- D. null
- E. null
NullPointerException

Answer: C

Q> Predict the nature of the following code

```
class Test {  
    public static void main(String[] args) {  
        int[] a; //local variable -> can't be used without initialization  
        System.out.println(a);  
        System.out.println(a[0]);  
    }  
}
```

- A. Compile Time Error
- B. 0
- C. [I@...
0
- D. null
- E. null
NullPointerException

Answer: A

